DeePromClass: Delineator for Eukaryotic Core Promoters Employing Deep Neural Networks

Hemanth Kari, Surya Manikhanta Sowri Bandi, Aditya Kumar, and Venkata Rajesh Yella

Abstract—Computational promoter identification in eukaryotes is a classical biological problem that should be refurbished with the availability of an avalanche of experimental data and emerging deep learning technologies. The current knowledge indicates that eukaryotic core promoters display multifarious signals such as TATA-Box, Inr element, TCT, and Pause-button, etc., and structural motifs such as G-quadruplexes. In the present study, we combined the power of deep learning with a plethora of promoter motifs to delineate promoter and non-promoters gleaned from the statistical properties of DNA sequence arrangement. To this end, we implemented convolutional neural network (CNN) and long short-term memory (LSTM) recurrent neural network architecture for five model systems with [-100 to +50] segments relative to the transcription start site being the core promoter. Unlike previous state-of-the-art tools, which furnish a binary decision of promoter or non-promoter, we classify a chunk of 151mer sequence into a promoter along with the consensus signal type or a non-promoter. The combined CNN-LSTM model; we call "DeePromClass", achieved testing accuracy of 90.6%, 93.6%, 91.8%, 86.5%, and 84.0% *for S. cerevisiae*, *C. elegans*, *D. melanogaster*, *Mus musculus*, and *Homo sapiens* respectively. In total, our tool provides an insightful update on next-generation promoter prediction tools for promoter biologists.

Index Terms—Convolutional neural network (CNN), core promoter, deep learning, eukaryotic Promoter Prediction, long short-term memory (LSTM)

1 Introduction

DNA makes up the source code of life, and the spatiotemporal gene expression is undertaken by the transcription program. A core promoter is a small segment of DNA that acts as the gateway to transcription [1]. It encompasses the transcription start sites (TSS) and functions as recognition sites for the transcription initiation machinery. In eukaryote core promoter, often characterized by a plethora of cognate sequence motifs, in particular, TATA box, Inr element (initiator element), BRE (TFIIB recognition elements BREu and BREd), DPE (downstream promoter element), and pause button, etc. precisely located relative to the transcription start sites [2], [3], [4], [5], [6], [7]. They also contain unique structural signals and non-canonical promoter structural motifs, namely, low stability regions, A-tracts, and non-B-DNA motifs [2], [8], [9], [10], [11], [12], [13]. Of note, these elements exist in a combinatorial fashion or independent manner or may not present at all in a subset of genes [3], [7], [14]. However, evolutionarily conserved transcriptional machinery recognizes the unique arrangement of sequences of the core promoter region strenuously and initiates the transcription of the template strand [15].

Marking promoter sequences in the genome are one of the classical and challenging tasks in biology. Promoter identification in genomes is accomplished with different goals such as deciphering gene expression, regulatory logic, annotation of genomes, depicting the statistical properties of non-coding genome, synthesis of novel promoters and learning the evolution of *cis*-regulatory architecture. Experimental setups for promoter characterization include localizing transcription start sites, mapping DNA-transcription factor interactions, transcription bubble snapshots, etc in the complex maze of genomes [16]. Although these approaches are exact, often time-consuming, cumbersome, exorbitant, and labour-exhaustive. On the other hand, computational methods deliver reliable results within a short period when the genomic sequence information is available. These tools for screening promoters utilize the presumption that promoter sequences differ in their nucleotide statistical properties in comparison to other genomic segments [16]. Promoter prediction methods can be broadly categorized into "search by signal (consensus motif information)", "search by content (*k*-mer information)", "search by structure (DNA structural descriptors such as duplex stability)", "integrated (combination of two or more features)", "hybrid (sequence information and other experimental information)" and "homology-based methods (orthologous gene information)" [16]. These features can be integrated with statistical learning models such as machine learning to improve performance. Over the past two decades, several promoter prediction algorithms have been developed utilizing the classical machine learning approaches, viz, artificial neural networks, Naive Bayes, principal component analysis, random forest classifier, relevance vector machine, and support vector machine, etc. (reviewed in [17], [18], [19]). With the accessibility of high-throughput genomic experimental data and the emergence of several novel algorithms, recent studies employed deep learning strategies for promoter prediction in eukaryotes.

Deep learning is a subgroup of machine learning techniques, acclaimed for its unprecedented predictive performance. The concept is derived from the ideology of how a biological neuron of the intricate nervous system works. The different types of neural networks in deep learning are artificial neural networks (ANN), convolutional neural networks (CNN), and recurrent neural networks (RNN). CNN is most often applied to image processing problems in which the connectivity layout among neurons is inspired by the architecture of the animal visual cortex. Long short-term memory (LSTM) belongs to the recurrent neural network architecture. The chain-like framework of LSTM permits it to hold information for longer periods and can be implemented for various tasks that mostly include prediction based on earlier information. Few state-of-the-art deep learning-based eukaryotic promoter predictors have been developed in recent times namely CNNProm [20], Qian *et al.* improved CNN [21], DCDE-MSVM [22], DeeReCT-PromID [23], DePromoter [24], and Depicter [25]. CNNPorm is based on the convolution neural network model for the identification of TATA-containing and TATA-less promoters of *A. thaliana*, *M. musculus*, and *H. sapiens* [20]. Improved CNN Architecture for Promoter Recognition by Qian *et al.* was designed with a major focus on transcription factor binding sites (TFBSs) in the human promoter sequence by keeping non-element sequence compressed form while promoter element sequences uncompressed [21]. DCDE-MSVM utilizes the deep convolutional divergence encoding (DCDE)" followed by multiple support vector machines (MSVM) to categorize genomic sequences into promoter and non-promoter classes in the human genome [22]. DeeReCT-PromID implemented

- *Hemanth Kari and Venkata Rajesh Yella are with the Department of Biotechnology, Koneru Lakshmaiah Education Foundation, Guntur, Andhra Pradesh 522502, India. E-mail: karihemanth19@gmail.com, yvrajesh_bt@kluniversity.in.*
- *Surya Manikhanta Sowri Bandi is with the Department of computer science, Indian Institute of Information Technology, Sricity, Chittoor, Andhra Pradesh 517646, India. E-mail: suryamanikhantasowri.b17@iiits.in.*
- *Aditya Kumar is with the Department of Molecular and Biotechnology, Tezpur University, Tezpur, Assam 784028, India. E-mail: aditya@tezu.ernet.in.*

Manuscript received 23 May 2021; revised 4 February 2022; accepted 17 March 2022. Date of publication 30 March 2022; date of current version 3 February 2023.
The work of Hemanth Kari and Venkata Rajesh Yella was supported in part by the Department of Biotechnology, and the management of the Koneru Lakshmaiah Education Foundation. This work was supported by the Early-Career Research under Grant ECR/2017/001100/LS from the Science and Engineering Research Board (SERB), Department of Science and Technology (DST), Government of India.
(Corresponding Author: Venkata Rajesh Yella.)
Digital Object Identifier no. 10.1109/TCBB.2022.3163418

1545-5963 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
thorized licensed use limited to: K.L. University(F.K.A. Koneru Lakshmaiah College of Engineering). Downloaded on July 25,2023 at 10:42:46 UTC from IEEE Xplore. Restrictions app

TABLE 1
Promoter and Non-Promoter Data Used

Organism	Number of sequences	
	Positive dataset	Negative dataset
S. cerevisiae	5117	5117
C. elegans	6364	6364
D. melanogaster	16972	16972
Mus musculus	25111	25111
Homo sapiens	29598	29598

[-100 to +50] and [+100 to +250], relative to the transcription start sites for protein-coding genes are chosen as positive (promoter) and negative (non-promoter) datasets respectively. The sequence lengths of 151 nucleotides for both positive and negative datasets have been utilized for training and testing.

TABLE 2
Multifarious Promoter Consensus Sequence Signals Implemented in the DeePromClass

Promoter Sequence Consensus Motifs	
TATA-Box	*TATAWAWR*
Inr Human	*BBCABW*
Inr Fly	*TCAGTY*
TCT	*YYCTTTYY*
BREu	*SSRCGCC*
BREd	*RTDKKKK*
DCEI–DCEIII	*CTTC, CTGT, and AGC*
XCPE1	*DSGYGGRASM*
XCPE2	*VCYCRTTRCMY*
Pause-button	*KCGRWCG*
DPE	*RGWCGTG or RGWYVT or GCGWKCGGTTS*
MTE	*CSARCSSAACGS*
Sp1 element	*GGCGGG or GGGCGG*
DNA Structural Motifs	
G-quadruplex	$G_{3-5}N_{1-7}G_{3-5}N_{1-7}G_{3-5}N_{1-7}G_{3-5}$
i-motif	$C_{3-5}N_{1-7}C_{3-5}N_{1-7}C_{3-5}N_{1-7}C_{3-5}$
A-tract	A_7 *or* T_7
G-tract	G_7 *or* C_7

Signals include functional promoter sequence motifs and DNA structural motifs. Corresponding IUPAC symbol code is: S = C or G; W = A or T; R = A or G; Y = C or T; K = G or T; M = A or C; B = C or G or T; D = A or G or T; V = A or C or G.

an approach of transcription start site position recognition for longer promoter regions based on CNN for human genomes [23]. DeePromoter was designed by combining several convolution layers with bidirectional long short-term memory-based RNN for predicting TATA-containing and TATA-less promoters of *M. musculus* and *H. sapiens* [24]. The most recent method, the Depicter (Deep learning for predicting promoter), combines multiple convolutional neural networks and capsule layers for predicting TATA-containing, non-TATA and indistinguishable promoters of four species, *Drosophila melanogaster*, *Arabidopsis thaliana*, *Mus musculus*, and *Homo sapiens* [25]. All these methods were either developed for picking up TATA-containing and TATA-less promoters or based on longer promoter regions of size >251 nucleotides. According to recent updates, the majority of functional core promoter elements are limited to -40 to +40 region relative to the TSS and structural signals can be positioned in -100 to +50 region [1], [2], [5], [12], [26], [27]. Of note, the core promoter architecture is very complex with several promoter signals, context, and DNA structural properties playing a role. Further, biologists are often interested in recognizing the core promoter fragments along with underlying signature motifs in the genome. Inscripting these limitations of previous state-of-the-art tools, we formulated a novel promoter prediction tool "DeePromClass (Deep learning-based Promoter Classifier)" by employing combined architecture of several layers of CNNs with LSTM and regular expression search strategy of promoter signals for five eukaryotic species *S. cerevisiae*, *C. elegans*, *D. melanogaster*, *Mus musculus*, and *Homo sapiens* respectively.

2 Data Sets

2.1 Promoter Dataset Collection

Our Datasets encompass promoter and non-promoter datasets for five eukaryotic model species *Saccharomyces cerevisiae*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Mus musculus*, and *Homo sapiens*. Promoter data and non-promoter data are regarded as positive and negative data for training the deep learning model. We gathered positive and negative datasets from Eukaryotic Promoter Database (EPD) that provides experimentally screened transcription start site information for eukaryotes [28]. EPD is an excellent resource for the advancement of computational research in promoter biology. The EPDnew webpage (https://epd.epfl.ch/EPDnew_database.php) has been accessed to extract the promoter data. Using the EPD select/Download tool, we retrieved the promoter and non-promoter sequence information. The sequence length of 151 nucleotides[-100 to +50] relative to transcription start sites has been chosen as positive data for this analysis. The -100 to +50 region indicates the highest core promoter activity with the majority of the promoter sequence signals are located within -40 to +40 [27], [28]. For the negative data set, the sequences from [+100 to +250] relative to TSS were extracted. The region [+100 to +250] often falls into the coding region. Of note, it is essential to screen promoter sequences from the coding region which lies in the immediate vicinity. The numbers of sequences extracted are listed in Table 1. For comparative analysis, same numbers of sequences for regions, [-40 to +40], [+100 to +180], [-250 to +50] and [+100 to +250] from EPD. [-40 to +40] and [-250 to +50] are regarded positive data while [+100 to +180] and [+100 to +250] noted to be negative dataset. The positive (promoter) and negative (non-promoter) datasets are labeled, concatenated, shuffled, and transformed to the required input format for compiling deep learning models. The transformed datasets were split with a ratio of 80:20 to make up the training and testing datasets for model development. Validation of the tool has been carried out using 20% of the training dataset.

2.2 Promoter Sequence Motifs

Multifarious promoter consensus sequence signals TATA-Box, initiator element (Inr Human and Inr Fly), TCT-element, TFIIB recognition elements (BREu and BREd), Downstream core promoter elements (DCEI–DCEIII), motif ten element (MTE), X Core Promoter Element (XCPE1 and XCPE2), Pause-button, downstream promoter element (DPE) and sp1 element and DNA structural motifs A-tracts, i-motifs, and G-quadruplex motifs have been implemented in the tool as a regular expression search strategy. Promoter sequence information has been taken from a recent study that compiled experimentally validated consensus motifs [6]. DNA structural motif search criterion was implemented as in our previous work [29]. The motif list along with consensus sequences was provided in Table 2.

2.3 Tools Used for Design of Model

DeePromClass has been implemented in python 3.0 using TensorFlow (tensorflow-gpu) libraries (https://www.tensorflow.org/). TensorFlow is the most widely used open-source machine learning framework with major stress on deep neural networks [30]. Higher-order neural network libraries from Keras were chosen to run on top of the TensorFlow framework. The python tools matplotlib (https://matplotlib.org/), numpy [31] (https://numpy.org/), and pandas [32] (https://pandas.pydata.org/) were employed for data processing, manipulation, visualization, and statistical analysis. The Scikit-learn predictive data analysis tool (https://scikit-learn.org/stable/)

thorized licensed use limited to: K.L. University(F.K.A. Koneru Lakshmaiah College of Engineering). Downloaded on July 25,2023 at 10:42:46 UTC from IEEE Xplore. Restrictions app

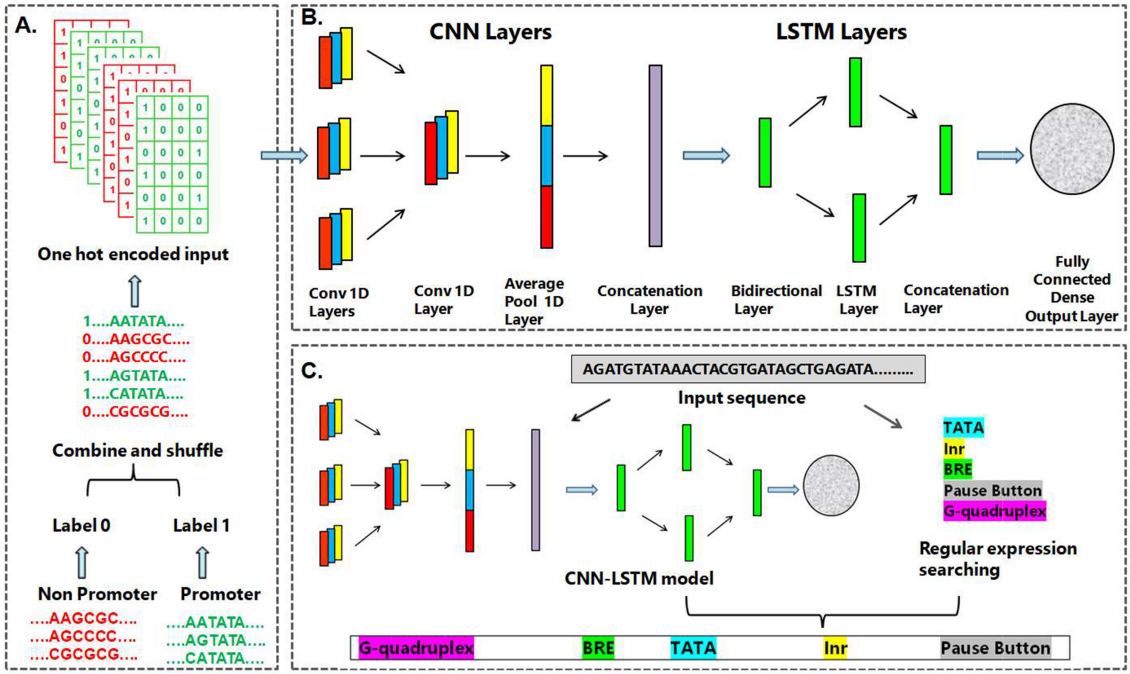

Fig. 1. The architecture of DeePromClass. (A) The promoter and non-promoter datasets are labeled, concatenated, shuffled, and one-hot encoded. (B) The one-hot encoded input layer is connected to convolutional layers and the output is then connected to LSTM layers. The end LSTM layers are then connected to the dense layer for classification. (C) An input sequence is tested through the CNN-LSTM module and regular expression search module to classify non-promoter or promoter with the presence of signature motifs.

has been used for computations such as precision, recall, F-score, accuracy, and receiver operating characteristic (ROC) metric [33].

2.4 Building Sequence Promoter Class Model With Deep Learning

The affirmed DeePromClass architecture involved several layers of one-dimensional convolutional neural networks with numbers of layers varying for species is combined with a long short-term memory framework layer. The outline of the architecture is displayed in Fig. 1.

2.4.1 Sequence Encoding

The DNA data has "A, T, G, C" nucleotides which cannot be interpreted by the machine for modeling. Hence, we have converted the dataset containing A, T, G, and C into a numerical vector as 1, 2, 3, and 4 respectively. Then, using the "to_catergorical" function from the Keras application programming interface the data has been one-hot encoded which consists of only 0 and 1. The one-hot encoded training dataset labeled for the promoter and non-promoter sequence was loaded as the input layer of the initial CNN model.

2.4.2 Convolutional Neural Network Layer

The CNN design comprises six (may vary for each *species*) one-dimensional convolutional layers. All the six convolutional layers are supervened by batch normalization. The second, fourth and sixth batch normalization steps are followed by average pooling and drop-out layers with a pool size of 2 and drop-out value of 0.5, respectively. The first two convolution layers were implemented with a filter size of 32; the third and fourth convolution layers use filter size of 64 and the final two convolution layers use filter size of 124. Kernel size of five is used for the first and second layer while 3 is used for the other four layers with the smaller kernel size of 5 with a common stride of value = 1. Non-linearity to all six CNNs has been added using the activation function, Rectified Linear Unit. The overfitting problem is controlled using the L1L2 kernel regulizer (which is the sum of the absolute and squared weights).

2.4.3 Long Short-Term Memory Layer

The extracted output features from the six CNN layers are pooled, flattened, and augmented through a bidirectional and long short-term memory layer with an output dimensionality of 1024. The network was further deepened using two consecutive layers with ReLU activation and the final Softmax activation methods respectively to arrive at the final model. Finally, we compiled the model with a categorical accuracy metric, adam optimization algorithm, and categorical_crossentropy loss function. The loss function used in the model is a categorical cross-entropy function to account for the stochasticity of the algorithm. Furthermore, we have used early stopping recall methods to stop the model from getting overfit (Supplementary Fig. 1, which can be found on the Computer Society Digital Library at http://doi.ieeecomputersociety.org/10.1109/TCBB.2022.3163418).

2.4.3 Regular Expression Module

Outside of the deep learning cascade, a regular expression search module with a multitude of sequence motifs was constructed (Table 2).

2.5 Performance Measures

To evaluate the performance of DeePromClass and to provide a comparison with other methods, we computed precision (positive predictive value), recall (true positive rate or sensitivity), F-score (harmonic mean of recall and precision), and accuracy [34] as follows

$$\text{Precision} = \frac{\text{True Positive (TP)}}{\text{True Positive (TP)} + \text{False Positive (FP)}}$$

$$\text{Recall} = \frac{\text{True Positive (TP)}}{\text{True Positive (TP)} + \text{False Negative (FN)}}$$

$$\text{F} - \text{Score} = \frac{\text{2x Precision x Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

thorized licensed use limited to: K.L. University(F.K.A. Koneru Lakshmaiah College of Engineering). Downloaded on July 25,2023 at 10:42:46 UTC from IEEE Xplore. Restrictions app

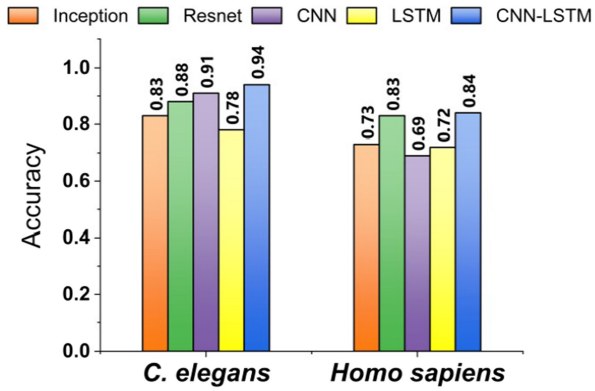

Fig. 2. Comparison of the performance of "DeePromClass" with other deep learning architectures. Bars indicate the testing accuracy for Inception, Resnet, CNN-only, LSTM-only, and CNN-LSTM (DeePromClass) models in *C. elegans* and *Homo sapiens*.

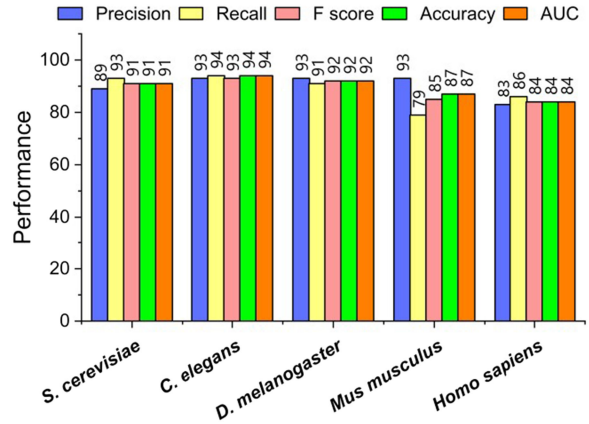

Fig. 3. Performance of DeePromClass on five different model systems. Performance measures namely precision, recall, F-score, testing accuracy, and Area under ROC curve (AUC) were computed for 151 nucleotide promoter and non-promoter datasets for five model organisms.

The performance metrics, precision, recall, F-score, and accuracy can account for the qualitative and quantitative performance of the tool. It should be noted that precision, recall, and F-scores are derived based on the positive and negative datasets used in this study.

3 Results and Discussion

Deep learning, is a blooming machine learning approach known for its unparalleled performance, and has now conspicuous in various fields of science and industry. The major objective of the current analysis is to design a new deep learning-based promoter prediction tool on the genomic scale to predict the putative promoters in a given genome sequence closely related to five model organisms *S. cerevisiae*, *C. elegans*, *Drosophila melanogaster*, *Mus musculus*, and *Homo sapiens* and characterize the sequence and structural motif signals in the predicted sequence. Despite the availability of excellent tools like CNNProm [20], DeePromoter [24], and Depicter [25] they do not consider a plethora of sequence motifs functionally conserved in the core promoter of eukaryotes. Here we proposed a different architecture for a smaller promoter region [-100 to +50] combining a simple regular expression strategy with a CNN-LSTM architecture. To this end, we extensively utilized the power of deep neural networks, knowledge of core promoter signals [4], [6], [29], and highly reliable promoter sequence data from the eukaryotic promoter database [28] and developed a promoter prediction tool we call "DeePromClass". We employed the TensorFlow robust machine learning system to enable large-scale promoter and non-promoter dataset training and inference [30]. It should be noted that not only the presence of sequence signals but the degenerated structural arrangement of the overall core promoter sequence is important for the orchestration of transcription initiation machinery [15], [35], [36], [37]. Hence, we separated the motif search as a separate outside layer from CNN-LSTM architecture. We validated the performance of DeePromClass using other *in-house* deep learning networks and compared with the state-of-the-art promoter prediction tools.

3.1 Comparison of DeePromClass Architecture With CNN, LSTM, Inception, and Resnet Architectures

To validate the performance of the DeePromClass network, we have compared the DeePromClass network with four sophisticated models, (1) Inception network, (2) Resnet network (3) Convolutional neural network-only model, and (4) LSTM neural network-only models respectively. An inception network was designed with 9 inception v1 modules with dimension reductions blocks and with each inception block containing 6 different layers of 1d convolutional layers. A Resnet model was created with each identity module having two 1d convolutional layers and each residual module having three 1d convolutional layers. For different sets of block layers, the connected network is further connected to the dense layer for classification. The network is 16 layers deep with each identity module connected to a residual module. The CNN model consists of only 8 convolutional layers connected to each other followed by a flattened layer. While the LSTM network constitutes 8 bidirectional LSTM layers of 1024 units which are later connected to dense layers. The DeePromClass differs from two layers with the input layer being connected to convolutional layers and the output of convolutional layers is then connected to LSTM layers, which in turn connected to the dense layer for classification. The same input layer of size 151 is loaded to the five networks. We tested these models on *C. elegans* and human datasets. Testing accuracy values obtained for the Inception, Resnet, CNN, LSTM, and CNN-LSTM models for *C. elegans* datasets are 0.83, 0.88, 0.91, 0.78, and 0.94 and for human datasets, the values are 0.73, 0.83, 0.69, 0.72, and 0.84 respectively (Fig. 2, Supplementary Table 1, available online). It has been observed that the Convolutional-LSTM model has better performance (higher accuracy) than other models. Hence CNN-LSTM has been considered for training on other model systems. Receiver operator characteristic has also been implemented on 20 percentiles of the training test dataset (validation dataset) to validate the performance of the Convolutional-LSTM model. The performance of DeePromClass (precision, recall, F-score, and accuracy) is shown in Fig. 3 and Supplementary Table 1, available online. It is observed that DeePromClass achieved good performance in all five species studied. The DeePromClass achieved a testing accuracy of 90.6%, 93.6%, 91.8%, 86.5%, and 84.0% *for S. cerevisiae*, *C. elegans*, *D. melanogaster*, *Mus musculus*, and *Homo sapiens* respectively which is considered as excellent performance for a promoter prediction tool. Previous tools have not reported the deep learning models for *C. elegans* and *S. cerevisiae*.

3.2 Comparison of DeePromClass Architecture With State-of-Art Tools

Next, we attempted to compare the performance of our tool with previously published state-of-the-art tools *viz* CNNProm [20], Deepromoter [24], and Depicter [25]. Recent research by song *et al.* [38] compared the performance of 19 accessible promoter prediction tools for prokaryotes and eukaryotes with 58 benchmarked promoter and non-promoter datasets. Their results indicated that conventional machine learning and deep learning derived tools are superior in performance compared to signal, k-mer, or structure-

thorized licensed use limited to: K.L. University(F.K.A. Koneru Lakshmaiah College of Engineering). Downloaded on July 25,2023 at 10:42:46 UTC from IEEE Xplore. Restrictions app

TABLE 3
Comparison of Performance of DeePromClass with Other State-of-the-Art Tools

Organism		TATA		NON-TATA	
		Pr	Re	Pr	Re
Drosophila	DeePromClass	0.96	0.98	**0.92**	0.95
	Depicter	1	-	0.90	-
Homo sapiens	DeePromClass	**0.99**	**0.98**	0.93	0.80
	DePromoter	0.93	0.95	0.97	0.95
	CNNProm	0.75	0.91	0.58	0.83
	Depicter	0.98	-	0.89	-
Mus musculus	DeePromClass	0.97	0.98	0.96	0.86
	DePromoter	0.92	0.95	0.91	0.90
	CNNProm	0.68	0.96	0.54	0.86
	Depicter	1	-	0.97	-

Benchmark datasets utilized for Depictor have been used to compare the performance of DeePromClass. It should be noted that CNNProm uses 251 nucleotide datasets. Code: Pr- precision, and Re-recall.

based tools. Support vector-based method iProEP [39], deep learning-based methods CNNProm, Deepromoter, and Depicter are found to be good predictor. The Author's highlight that Depicter being the superior promoter predictor. However, the performance values of the tools vary in a species-specific manner [38]. The actual region chosen for training the model is the [-249 to +50] region for Deepromoter and Depicter, whereas we used the [-100 to +50] region. Hence, we trained our model with 300 nucleotide datasets utilized by depictor tool and compared with the reported performance to their published literature. However, the comparison cannot be carried out straightforwardly as the choice of species used for building models vary for each tool. The reported precision and recall values were listed as Table 3. CNNProm, DeePromoter, and Depicter mainly concentrated on the TATA and non-TATA promoter classification of *Homo sapiens*, and *Mus musculus promoters*, hence we trained DeePromClass separately for two classes. Results indicated that the performance of DeePromClass, Depictor and Deepromoter are comparable with one tool outperform the other in specific species or promoter type manner. Interestingly, DeePromClass is superior to other tools for human TATA and Drosophila Non-TATA datasets (Table 3). It should be noted that The CNN-LSTM architecture may seem similar to that of DePromoter except for the fact that the architecture varies a lot with varying number of layers of CNN and LSTM's for each organism model, modeling of the data and dataset itself varied. There is no data augmentation involved in DeePromClass. DePromoter took the sequence length of 300bp and subdivided it into parts to increase the dataset, whereas DeePromClass used complete sequence length without any subdivision. We also tested our tool with datasets used by traditional machine learning tool iProEP for human datasets and observed that iProEP provided superior performance (sensitivity = 92.3) compared to our tool (89%). However, compared to all recent state-of-the-art tools, the main advantages of our tool is along with classifying a given sequence as a promoter with a confidence score; we also provide biological insights with a plethora of functional motifs that are also embedded in the predicted promoter regions. The promoter is the arrangement of sequences and combination of promoter elements.

3.3 DeePromClass Application

The DeePromClass first takes the input sequence and classifies the input sequence if it is a promoter sequence or a non-promoter sequence based on the "promoter confidence score in percentage". Furthermore, it lists the name of the core promoter elements present in the identified sequence. Currently, the DeePromClass tool is provided as a source code form through Github static hosting service.

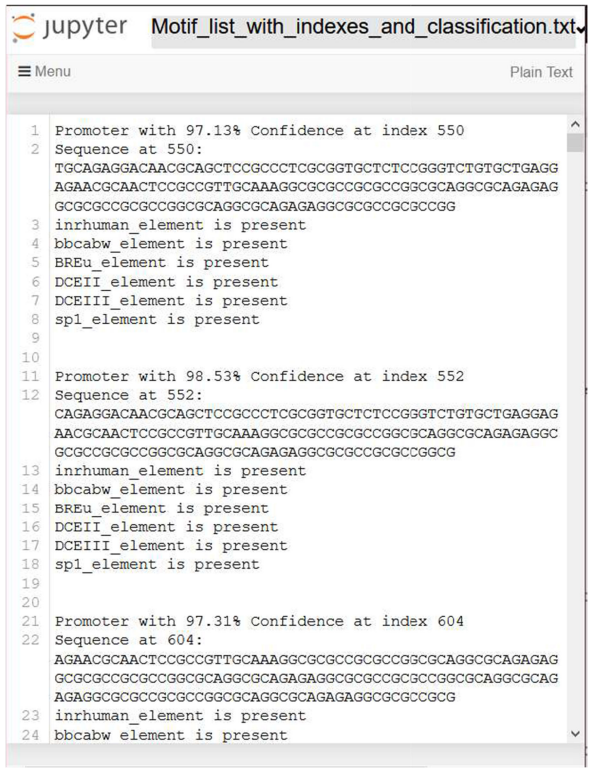

Fig. 4. Final Output of promoters predicted along with indexes and classification. DeePromClass tool can be implemented through an interactive python notebook such as jupyter or Googlecolab notebook with the installation of module requirements. The figure displays the output of the result text file saved in the local system.

We provide the compiled models for the five *species*, *C. elegans*, *D. melanogaster*, *Homo sapiens*, *Mus musculus*, and *S. cerevisiae* along with the main "promoter finder and classifier" script. The user is required to download the toolkit and should run the main script through an interactive python notebook such as jupyter or Googlecolab notebook with the installation of module requirements. The test DNA sequence should be provided to the tool as "Input_data.txt" in Fasta format. There is no limit on the size of the sequence. The tool outputs the list of the promoters with a size of 151 fragments into a separate file along with indexes of each of the found possible promoters. These promoters are also classified based on the consensus signals found in the 151 regions. A score of percentage confidence was provided along with the start index of promoters. The percentage confidence range 95-100 closer to 100 is regarded as highly reliable. A score lower than 95 may not be a promoter. Promoter sequences without any motif listed in the tool are predicted as "unable to classify". A clear instruction note for employing the tool is provided as a readme.txt file. An example prediction output result of the tool is shown in Fig. 4. The source code of DeePromClass and the associated datasets are freely available at https://github.com/hemanthkari/DeePromClass.git.

4 Conclusion

Delineation of promoter classes is crucial for understanding the evolution of promoter sequences and diversity in *cis*-regulatory logic. In the current study, we constructed a novel promoter delineator tool, DeePromClass, inspired by the recent prosperity of deep learning in various fields of research. We applied one-dimensional convolutional neural networks (1D CNN) and recurrent neural networks (LSTM) to train promoter and non-promoter datasets and integrated the regular expression search strategy of several promoter elements to output promoter prediction results. DeePromClass performance is on par with previously published deep learning-based tools. However, the main limitation of the tool is slow for whole genomic sequences at

thorized licensed use limited to: K.L. University(F.K.A. Koneru Lakshmaiah College of Engineering). Downloaded on July 25,2023 at 10:42:46 UTC from IEEE Xplore. Restrictions app

this point due to the computational complexity of the defined problem with the current approach. Some models employed here perform better on some organisms. Furthermore, our tool may be needed to be tested with hyperparameter tuning and rigorous training sessions on other model species for more applicability for a biologist. However, our algorithm is scalable, tunable, and unique in its kind to delineate promoter classes, experts can improvise the accuracy of the model by changing the number of layers and feature mapping and implementing new promoter signals. We believe that our work will enhance newer and better algorithms that integrate the DNA one-hot encoding with a series of experimentally determined motif data and structural information. It may help promoter synthetic biology to design better promoters. Henceforth, we will improvise the tool by combining a multitude of signal, structure, and contextual features along with training on several other model systems with web server implementation.

REFERENCES

[1] T. Juven-Gershon, J.-Y. Hsu, J. W. Theisen, and J. T. Kadonaga, "The RNA polymerase II core promoter—The gateway to transcription," *Curr. Opin. Cell Biol.*, vol. 20, no. 3, pp. 253–259, 2008.

[2] V. R. Yella and M. Bansal, "DNA structural features of eukaryotic TATA-containing and TATA-less promoters," *FEBS Open Bio*, vol. 7, no. 3, pp. 324–334, Mar. 2017.

[3] S. Weingarten-Gabbay and E. Segal, "The grammar of transcriptional regulation," *Hum. Genet.*, vol. 133, no. 6, pp. 701–711, Jun. 2014.

[4] A. Sloutskin *et al.*, "ElemeNT: A computational tool for detecting core promoter elements," *Transcription*, vol. 6, no. 3, pp. 41–50, 2015.

[5] M. Levo and E. Segal, "In pursuit of design principles of regulatory sequences," *Nature Rev. Genet.*, vol. 15, no. 7, pp. 453–468, Jul. 2014.

[6] V. Haberle and A. Stark, "Eukaryotic core promoters and the functional basis of transcription initiation," *Nature Rev. Mol. Cell Biol.*, vol. 19, no. 10, pp. 621–637, Oct. 2018.

[7] Y. M. Danino, D. Even, D. Ideses, and T. Juven-Gershon, "The core promoter: At the heart of gene expression," *Biochimica Biophysica Acta*, vol. 1849, no. 8, pp. 1116–1131, Aug. 2015.

[8] M. Bansal, A. Kumar, and V. R. Yella, "Role of DNA sequence based structural features of promoters in transcription initiation and gene expression," *Curr. Opin. Struct. Biol.*, vol. 25, pp. 77–85, 2014.

[9] J. Spiegel, S. Adhikari, and S. Balasubramanian, "The structure and function of DNA G-Quadruplexes," *Trends Chem.*, vol. 2, no. 2, pp. 123–136, Feb 2020.

[10] K. Szlachta, R. G. Thys, N. D. Atkin, L. C. T. Pierce, S. Bekiranov, and Y. H. Wang, "Alternative DNA secondary structure formation affects RNA polymerase II promoter-proximal pausing in human," *Genome Biol.*, vol. 19, no. 1, Jul. 2018, Art. no. 89.

[11] Q. Xiang, "Functional implications of local DNA structures in regulatory motifs," *Sci. World J.*, vol. 2013, 2013, Art. no. 965752.

[12] V. R. Yella and M. Bansal, "DNA structural features and architecture of promoter regions play a role in gene responsiveness of s. cerevisiae," *J. Bioinf. Comput. Biol.*, vol. 11, no. 6, Dec. 2013, Art. no. 1343001.

[13] V. R. Yella, A. Kumar, and M. Bansal, "Identification of putative promoters in 48 eukaryotic genomes on the basis of DNA free energy," *Sci. Rep.*, vol. 8, no. 1, Mar. 2018, Art. no. 4520.

[14] A. L. Roy and D. S. Singer, "Core promoters in transcription: Old problem, new insights," *Trends Biochem. Sci.*, vol. 40, no. 3, pp. 165–161–71, Mar. 2015.

[15] X. Chen *et al.*, "Structural insights into preinitiation complex assembly on core promoters," *Science*, vol. 372, no. 6541, Apr. 2021, Art. no. eaba8490.

[16] V. R. Yella and M. Bansal, "In silico identification of eukaryotic promoters," in *Systems and Synthetic Biology*, Berlin, Germany: Springer, 2015, pp. 63–75.

[17] S. Singh, S. Kaur, and N. Goel, "A review of computational intelligence methods for eukaryotic promoter prediction," *Nucleosides Nucleotides Nucleic Acids*, vol. 34, no. 7, pp. 449–462, 2015.

[18] V. R. Yella, A. Kumar, and M. Bansal, "DNA structure and promoter engineering," in *Systems and Synthetic Biology*, Berlin, Germany: Springer, 2015, pp. 241–254.

[19] J. Zeng, S. Zhu, and H. Yan, "Towards accurate human promoter recognition: A review of currently used sequence features and classification methods," *Brief. Bioinform.*, vol. 10, no. 5, pp. 498–508, Sep 2009.

[20] R. K. Umarov and V. V. Solovyev, "Recognition of prokaryotic and eukaryotic promoters using convolutional deep learning neural networks," *PLoS One*, vol. 12, no. 2, 2017, Art. no. e0171410.

[21] Y. Qian, Y. Zhang, B. Guo, S. Ye, Y. Wu, and J. Zhang, "An improved promoter recognition model using convolutional neural network," in *Proc. IEEE 42nd Annu. Comput. Softw. Appl. Conf.*, 2018, pp. 471–476.

[22] W. Xu, L. Zhu, and D. S. Huang, "DCDE: An efficient deep convolutional divergence encoding method for human promoter recognition," *IEEE Trans. Nanobiosci.*, vol. 18, no. 2, pp. 136–145, Apr. 2019.

[23] R. Umarov, H. Kuwahara, Y. Li, X. Gao, V. Solovyev, and J. Hancock, "Promoter analysis and prediction in the human genome using sequence-based deep learning models," *Bioinformatics*, vol. 35, no. 16, pp. 2730–2737, 2019.

[24] M. Oubounyt, Z. Louadi, H. Tayara, and K. T. Chong, "DeePromoter: Robust promoter predictor using deep learning," *Front. Genet.*, vol. 10, 2019, Art. no. 286.

[25] Y. Zhu, F. Li, D. Xiang, T. Akutsu, J. Song, and C. Jia, "Computational identification of eukaryotic promoters based on cascaded deep capsule neural networks," *Brief. Bioinf.*, vol. 22, Jul. 2021, Art. no. bbaa299.

[26] T. Juven-Gershon and J. T. Kadonaga, "Regulation of gene expression via the core promoter and the basal transcriptional machinery," *Devlop. Biol.*, vol. 339, no. 2, pp. 225–229, Mar. 2010.

[27] S. Weingarten-Gabbay *et al.*, "Systematic interrogation of human promoters," *Genome Res.*, vol. 29, no. 2, pp. 171–183, Feb. 2019.

[28] R. Dreos, G. Ambrosini, R. C. Périer, and P. Bucher, "The eukaryotic promoter database: Expansion of EPDnew and new promoter analysis tools," *Nucleic Acids Res.*, vol. 43, no. D1, pp. D92–D96, 2015.

[29] V. R. Yella, A. Vanaja, U. Kulandaivelu, and A. Kumar, "Delving into eukaryotic origins of replication using DNA structural features," *ACS Omega*, vol. 5, no. 23, pp. 13601–13611, Jun. 2020.

[30] M. Abadi *et al.*, "Tensorflow: A system for large-scale machine learning," in *Proc. 12th USENIX Symp. Oper. Syst. Des. Implementation*, 2016, pp. 265–283.

[31] P. F. Dubois, K. Hinsen, and J. Hugunin, "Numerical python," *Comput. Phys.*, vol. 10, no. 3, pp. 262–267, 1996.

[32] W. McKinney, "Pandas: A foundational python library for data analysis and statistics," *Python High Perform. Sci. Comput.*, vol. 14, no. 9, pp. 1–9, 2011.

[33] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825−2830, 2011.

[34] Y. Jiao and P. Du, "Performance measures in evaluating machine learning based bioinformatics predictors for classifications," *Quantitative Biol.*, vol. 4, no. 4, pp. 320–330, 2016.

[35] V. Brazda, M. Bartas, and R. P. Bowater, "Evolution of diverse strategies for promoter regulation," *Trends Genet.*, vol. 37, pp. 730–744, Apr. 2021.

[36] J. Zrimec *et al.*, "Deep learning suggests that gene expression is encoded in all parts of a co-evolving interacting gene regulatory structure," *Nature Commun.*, vol. 11, no. 1, Dec. 2020, Art. no. 6141.

[37] C. Qiu *et al.*, "Universal promoter scanning by pol II during transcription initiation in saccharomyces cerevisiae," *Genome Biol.*, vol. 21, no. 1, Jun. 2020, Art. no. 132.

[38] M. Zhang *et al.*, "Critical assessment of computational tools for prokaryotic and eukaryotic promoter prediction," *Brief. Bioinf.*, vol. 23, 2021, Art. no. bbab551.

[39] H.-Y. Lai *et al.*, "iProEP: A computational predictor for predicting promoter," *Mol. Ther.-Nucleic Acids*, vol. 17, pp. 337–346, 2019.

▷ **For more information on this or any other computing topic, please visit our Digital Library at** www.computer.org/csdl.

thorized licensed use limited to: K.L. University(F.K.A. Koneru Lakshmaiah College of Engineering). Downloaded on July 25,2023 at 10:42:46 UTC from IEEE Xplore. Restrictions app